PISCINE MAKERS · EMBEDDING LAB

Semantic Agent Engine

A memory that knows what was decided,
and what is still true.

Architecture v4 · Jonathan Chayna · October 2026

Architecture v4 — seven named steps, from the arriving message to the cited answer. Every step has an input, an output, a component and a measure.

The idea in one minute

A classic search finds sentences that look alike, not necessarily the one that is still true. Here the message stays intact, the **chunker** extracts sentences with offsets, the **tagger** reads kind, status, point and value, the **linker** says what replaces what, **memory** keeps the current state and the whole history, the **retriever** blends words, meaning and graph, and the **answerer** replies with the source sentence — or refuses.

It understands

A proposal is not a decision. Status is explicit: proposed, confirmed, cancelled, completed.

It remembers

Message, sentence, speaker, date and proof keep their identity. Nothing is rewritten.

It does not guess

No evidence: “I don't know”. No invented name, number, date or recipient.

The seven steps, one line each

- **store** — the message arrives as-is; the database refuses updates
- **chunker** — sentences and units, exact offsets, short confirmations attached
- **tagger** — kind, status, intent, point, value — fine-tuned Laya
- **linker** — same point, different value, settled turn → supersedes link
- **memory** — append-only state; current = latest confirmed; the old stays
- **retriever** — BM25 + dense + graph, fusion, Qwen3-Reranker-0.6B
- **answerer** — context + state → LLM; cited answer or refusal, numbers checked

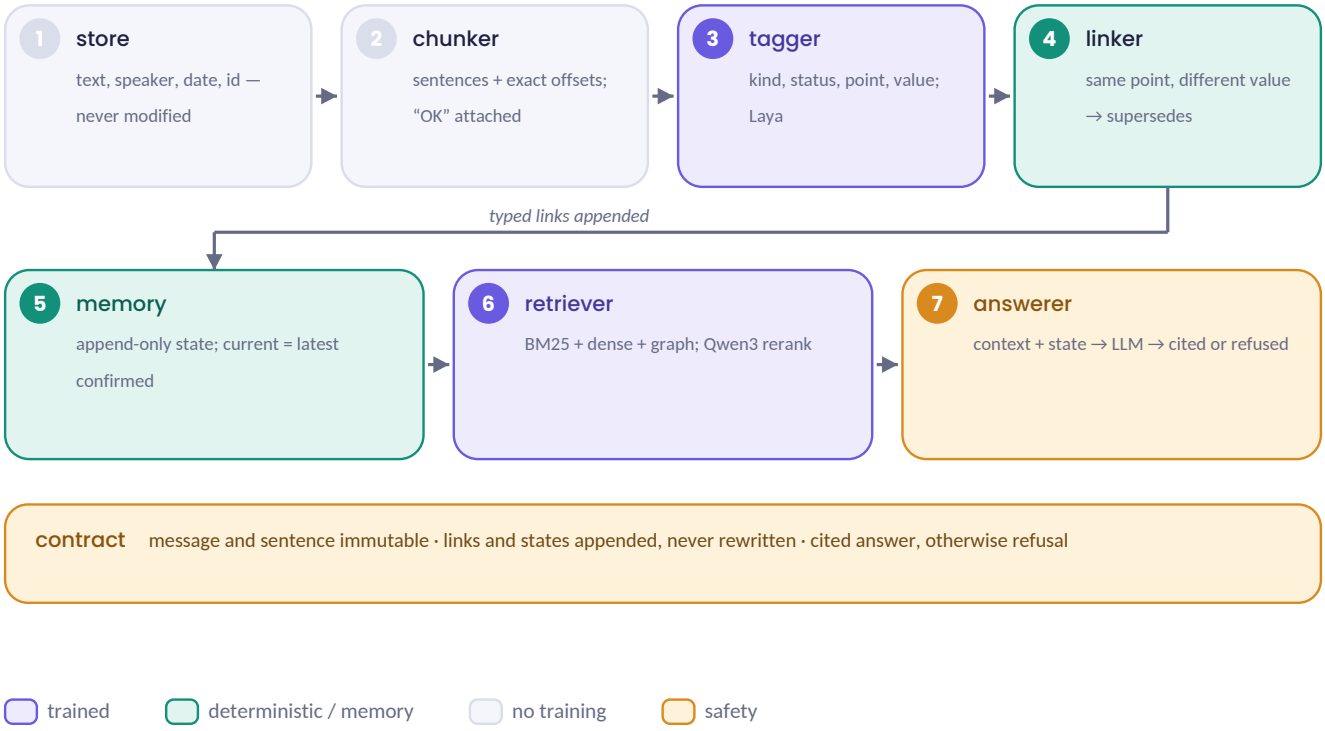
Iron rule: message and sentence are immutable; links and states are appended, never rewritten. Current state is a view, not a source of truth.

The number

0.778 correct answers over 5 hand-written meetings and 36 questions; the right sentence is in the top 10 for **96%** of them; **7** of 10 tricky questions are refused. Regenerated by **make meetings**.

The diagram

Seven named components, one path: write on the left, read on the right



- **Top row** — what arrives: the message is stored, split, tagged, then linked to the turns that carried the old value.
- **Bottom row** — what is read: memory holds the current state, the retriever ranks the turns, the answerer writes or refuses. The retriever never modifies memory.
- **The connector** between linker and memory is the only state-writing path: nothing else changes the current state.

Colours: purple = trained part, teal = deterministic or memory, grey = no training, amber = contract and refusal. Two steps deliberately have no training at all: the linker and memory.

The contract, in plain words

A message never changes; a sentence keeps its offsets; a replacement is an appended link; a state is an appended event; an answer cites a sentence or refuses. That is what makes the system auditable: you can always walk back from the answer to the exact text.

The path of a decision

"we're going with 4,500, it's agreed" → **store** (t:17, 11/03) → **chunker** (1 unit) → **tagger** (point = price, value = 4,500, confirmed) → **linker** (replaces the 4,800 of 03/03) → **memory** (current = 4,500, the 4,800 stays) → **retriever** (top 1) → **answerer** ("4,500, confirmed [traiteur:17:u0]").

The path of a question without a state

"how much is the caterer?" → **retriever** (the price turn comes up) → no state attached to that unit → **answerer** cites the turn directly ("4,500 [traiteur:17:u0]"). This is the case that failed before: a true fact with no state was not answerable.

1 On the example

One message, two pieces, two proofs

> *“For project X, we're pushing the launch to December. By the way, where are we on the back end?”*

- **Piece A** — “we're pushing the launch to December”: decision. The message stays intact; a **StateEvent** sets the launch to December, the old value stays in history.
- **Piece B** — “where are we on the back end?”: question. We search after storing A; the answer cites the source sentence.
- **A very short “OK, approved”** stays attached to the preceding message; it does not become an isolated sentence.

2 2 · Chunker — split without losing the proof

Documents

Neighbouring sentences about the same thing are grouped. Five sentences on one subject: one piece. One sentence about a car then one about a ball: two pieces.

Conversations

We cut as soon as the topic or intent changes, even mid-turn. A short “OK, approved” stays attached to the preceding message.

Code

One function = one piece. Simple code-reading rules, no artificial intelligence.

Always: even when a piece groups several sentences, each original sentence keeps its offsets and remains the **proof**. **text[start:end] == span.text** on all 103 turns: 0 mismatch.

Case	Rule	Example
two sentences, same topic	a single piece	budget proposed then approved
topic or intent changes	we cut	decision then question
a short “OK, approved”	attached to the previous message	it does not become a unit
code	one function = one piece	reading rules, no AI

The immutable contract

Record	What never changes
Message	message_id, source, project, speaker, timestamp, text, hash
Chunk	chunk_id, message_id, start, end, text, sentence_index
Unit	unit_id, message_id, project, slot, point, value, status, kind, evidence
LinkEvent	appended: from, to, relation, confidence, evidence
StateEvent	appended: event (project:slot:point), value, status, evidence, date

The **point** is the missing discriminator: without it, “keys”, “fibre” and “lease” shared one date key. The key is **(project, slot, point)**.

3 The seven steps

Each has a name, an input, an output and a measure

1. **store**
   The message arrives as-is: id, speaker, date, text. The database refuses updates; re-ingesting the same message does nothing. 0 duplicates.
2. **chunker**
   Splits into sentences and units with exact offsets. Short confirmations stay attached. Output: chunks pointing at their text.
3. **tagger**
   Fine-tuned multilingual Laya reads each unit: kind, status, intent, point and value. Change recall 1.00; 32 of 32 values are values, not sentence fragments.
4. **linker**
   Deterministic rule: same project, slot and point, different value, settled new turn → a **supersedes** link to the latest turn carrying the old value. 4 of 4 replacements linked.
5. **memory**
   Append-only state: a proposal stays open, a confirmation updates the current view, the old value stays in history. 12 of 12 current values; 0 status leak between units.
6. **retriever**
   Entities and events, then BM25 + dense, graph expansion, rank fusion and Qwen3-Reranker-0.6B. hit@1 .577 · hit@3 .846 · hit@10 .962.
7. **answerer**
   The question, the retrieved turns (id, date, speaker) and the current state go to the LLM; the answer is kept only if it cites a supplied excerpt and its numbers appear there — otherwise refusal. Cites the right sentence .808; 7 of 10 tricky questions refused.

What each step receives and returns

Step	Receives	Returns
store	raw message	immutable row
chunker	message	chunks + offsets
tagger	chunk	kind, point, value, status
linker	unit + older turns	supersedes link, or nothing
memory	links + states	current view + history
retriever	question	10 ranked turns
answerer	question + turns + state	cited answer or refusal

What changes from v2/v3

- **v2** had the right journey, but “replace” could rewrite the old message.
- **v3** made the meeting immutable, but current came from the latest message, not a confirmed state.
- **v4** names every step, adds offsets, the point, typed links and append-only state; current is the latest confirmed state, not the latest chatter.

4 6 · Retriever — find it, then read the state

The core stops here: the answer is a generation over this context

- **Interpret the query** — Intent, requested slot, people, dates; resolve “it”, “he”, “she”, “that”, “him”, “then”.
- **Entities and events** — Separate indexes: people, projects, objects, decisions, actions, events. “Pay him” finds the payment and the supplier.
- **BM25 + dense** — Exact words for names, dates, numbers; meaning for paraphrases; RRF fusion.
- **Graph expansion** — Traverse **same_event**, **answers**, **creates_action**, **supports**, **supersedes** to recover distant continuations.
- **Contextual reranking** — Qwen3-Reranker-0.6B rereads the question with each candidate; its ranking is fused with the pipeline's, not substituted for it.

The reranker, measured on 26 answerable questions

Ordering	hit@1	hit@3	hit@5
pipeline order	.346	.808	.846
reranker, default instruction	.500	.692	.846
reranker, French instruction	.423	.885	.885
fusion + French instruction	**.577**	**.846**	.885

Pure reranking wins top 1 and loses top 3; fusion keeps both. No fine-tune: 26 questions would be enough to win its own set and lose elsewhere.

The encoder choice, measured

Encoder	What we measured	Decision
e5-base	the reference on all 36 questions	shipped
jina v3	same level, sometimes ahead	nothing to gain
jina v5	same level as v3	nothing to gain
late chunking	clear gain when the query is contextual	kept behind --long-split

What retrieval still misses

- Five answerable questions whose point appears in no turn: “one now, one in Q3”, “the 3,000 gap”, “the cause of the incident”, two anaphoric “it's me” answers.
- Embeddings change nothing measurable: jina v3 and v5 tie with e5; e5-base ships for reproducibility. The gain came from late chunking, not the model.
- Late chunking only helps when the query is encoded with its context: index-time vectors with a context-free query invert the signal (9/40).

5 4 · Linker and 5 · Memory

Decide what replaces what, without ever rewriting

The linker's rule

- same **project**, same **slot**, same **point**
- a **different value** in a **new, settled** turn (not a question, not a hesitation)
- the link points at the latest turn carrying the old value
- nothing is rewritten: “replaced” is a link, not an edit

The state key

(project, slot, point) — the point separates “keys”, “fibre”, “lease”, “salary”, “deposit”.

A proposal stays **open**; confirmed / cancelled / completed updates the current view; the old value stays in history.

A unit only gets a status if it is that state's evidence: 76 leaks out of 103 units before the rule, **0** after.

Five real keys, exactly as memory writes them

Project	slot	point	Value	Status
nova	date	lancement	5 avril	confirmed
hiring	budget	salaire	52 000	confirmed
traiteur	budget	acompte	1 350	proposed
demenagement	date	clés	1er septembre	confirmed
incident	budget	utilisateurs	12 000	confirmed

What becomes “current”?

Unit	Status	Current?	Why
“we could raise it to 1,500”	proposed	no	a proposal never replaces a decision
“we approve 800”	confirmed	yes	confirmed, dated, with proof
“actually we cancel”	cancelled	yes (cancelled)	a cancellation is a state, not a silence
“he said he'd see”	unknown	no	no commitment made

One state, three messages

- **October 1** — “the budget is proposed at €1,200”: proposal, open state.
- **October 8** — “we approve €800”: confirmation; current becomes €800, €1,200 stays in history.
- **October 15** — “actually we cancel”: cancellation; current becomes cancelled, both values remain visible.
- **“We could raise it to €1,500” does not replace a confirmed €800.** It stays open.

Right away

- split, tag, index the message
- retrieve candidates from history
- append links and state, without rewriting

In the background

- rebuild entity/event indexes
- audit the reranker and thresholds
- propose retraining on corrected data

6 Results, step by step

5 hand-written meetings · 103 turns · 36 questions

Step	Measure	Result
1 · store	edit refused; re-ingest no-op	refused; 0 duplicates
2 · chunker	**text[start:end] == span.text**	0 mismatch on 103 turns
3 · tagger	change recall; values that are values	1.00; 32 of 32
4 · linker	replaced turn linked	4 of 4
5 · memory	current value; open proposal; leaks	12 of 12; 1 of 1; 0
6 · retriever	hit@1 / hit@3 / hit@10	.577 / .846 / .962
7 · answerer	cites the right sentence; tricky refused	.808; 7 of 10

End to end: 0.778 — formal 0.833, messy 0.750, tricky 0.700. Same set, three answerers:

Answerer	Correct	Cites right turn	Tricky refused
deterministic template	.694	.731	9/10
LLM alone (7B)	.583	.577	7/10
hybrid (shipped)	**.778**	**.808**	7/10
4B LLM	.278	0	10/10

The 4B refuses everything: on 8 GB, the 7B is the usable floor. Memory answers when it has the proof; the LLM only takes the rest.

7 What we train

Each model has a role and a measure; two steps have no training at all

Step	Model	Data	Measure
3 · tagger	Laya multilingue, fine-tuné	12,500 / 2,000 / 4,200 synthetic examples, family-disjoint	recall 1.00; 32/32
6 · retriever (meaning)	e5-base gelé	real + synthetic pairs	hit@10 .962
6 · retriever (rerank)	Qwen3-Reranker-0.6B, zéro-shot	none	+.231 hit@1 by fusion
4 · linker	deterministic rule	none	4/4
5 · memory	append-only rule	none	12/12; 0 leak
7 · answerer	Qwen2.5-7B-instruct + vérificateur	none	0.778 hybrid

Reranker fine-tuning was ruled out: 26 questions are enough to win its own set and lose elsewhere. The tagger has 18,700 examples and genuinely wins.

8 Measured limits

What the numbers say, unrounded

- **Synthetic only.** Five meetings, 36 questions, gold written by the author of the meetings. No held-out meeting: a solid baseline, not a leaderboard result.
- **The rules are fitted to these five meetings**: status, point, value. The next honest test is a meeting the rules have never seen.
- **hit@1 = .577.** The right sentence is in the top 3 for 85% of questions, in the top 10 for 96% — but not always first. The reranker wins top 1 and sometimes loses top 3; fusion limits the damage.
- **3 of 10 tricky questions still get an answer.** The 7B concludes from an intention that an action happened ("we'll send the offer" → "he accepted"). A larger model is the missing variable.
- **Two keys for one point.** A turn without a point writes **project:slot** next to **project:slot:point**. The linker should inherit the point when it is unambiguous.

What would change the verdict

- a held-out meeting that neither the rules nor the thresholds have seen
- a larger answerer (14B or hosted): the three remaining tricky answers are a model limit
- inherit the point when it is unambiguous, to remove the double key
- twenty more meetings to separate a rule that works from a rule that got lucky

9 Short glossary

Technical words in plain language

store	The original message, as-is: text, speaker, date, id. Never modified.
chunker	Splits the message into sentences and units; every sentence keeps exact offsets.
tagger	Reads each unit: kind, status, point, value. This is fine-tuned Laya.
linker	Rule that says which turn replaces which. It appends a link, it changes nothing.
memory	Current state is a view over appended events; history stays intact.
retriever	Exact words + meaning + graph, then a reranker: it finds candidate turns.
answerer	Question + retrieved turns + state → LLM. The answer cites or refuses; the verifier rejects any number absent from the proof.
abstention	Without enough evidence, the engine says "I don't know" instead of inventing.

Everything regenerates: **make meetings** (end to end), **make steps** (the table above), **make ask4** (one question, by hand).